

A ensemble method for industrial Anomaly detection and Localization

Jie Wang, Yanming Zhang, Ting Wang, YunLong Li, Jing Chen

Abstract. Currently, there are three main anomaly detection and localization algorithm, synthesis-base methods, reconstruction-based methods, embedding-based method. Different algorithms are suitable to different scenarios, and we are not limited to using one algorithm. On the contrary, we have integrated different algorithms and achieved optimal performance in practical.

Introduction

In this paper, we ensemble two different method, GLASS and INP-Former, and achieved optimal performance on MVTEC2 dataset. In addition, we have made improvements to these model, such as using better backbone, better feature fusion module, and more suitable training strategies. Next we will provide a detailed introduction to the specific content.

Method

Better backbone: In the original implementation, only weak backbones such as wrenset were used. In this paper, we use more modern transform based backbones and, if necessary, ensemble multiple backbones.

Better feature fusion module: In the original implementation, weak feature fusion modules such as Unfold and Pool were used. In this paper, we employed a multi-layer convolution network which have better performance.

Better training strategies: In this paper, we employed a better learning rate adjustment strategy and more precise weight decay control.

Main Results

The experimental results on the three AD datasets are presented in Tab. 1. On MVTEC-AD2 dataset, our method achieves SegF1 47.5 on private test dataset and SegF1 44.49 on private mixed test dataset.

Table 1. Single class anomaly detection performance on MVTEC-AD2

classes name	average	can	fabric	fruit jelly	rice	sheet metal	vial	wallplugs	walnuts
private	47.5	18.14	46.53	62.59	72.5	64.55	42.67	5.5	67.49
private mixed	44.49	10.13	31.74	62.05	73.74	62.32	43.96	5.06	67.21

